

Module -2

Step Response of First Order RC Circuits

Dr. Sathya P
Associate Professor
SENSE

Singularity Functions

- Singularity functions (also called switching functions) are functions that either are discontinuous or have discontinuous derivatives.
- The most widely used singularity functions in circuit analysis are
 - ❖ The Unit step function
 - ❖ The Unit Impulse function
 - ❖ The unit ramp function

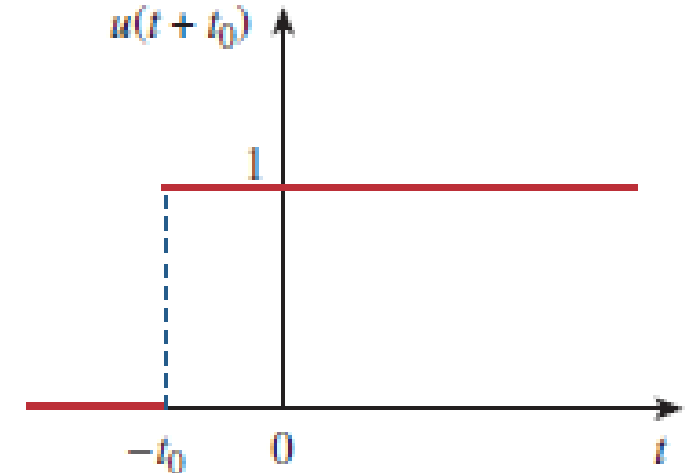
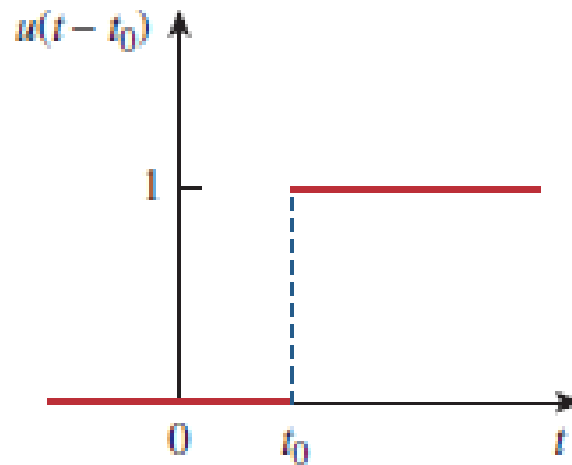
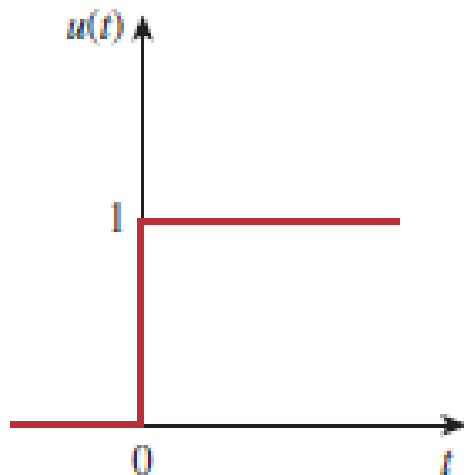
The Unit Step function

- The unit step function $u(t)$ is 0 for negative values of t and 1 for positive values of t .

$$u(t) = \begin{cases} 0, & t < 0 \\ 1, & t > 0 \end{cases}$$

$$u(t - t_0) = \begin{cases} 0, & t < t_0 \\ 1, & t > t_0 \end{cases}$$

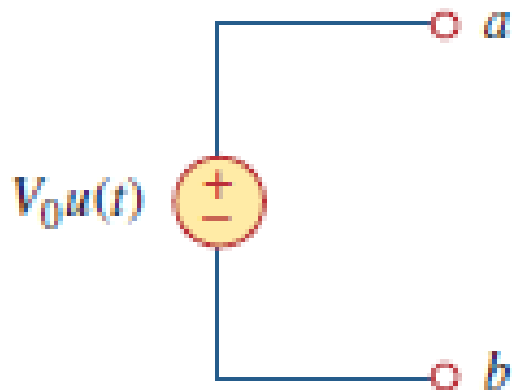
$$u(t + t_0) = \begin{cases} 0, & t < -t_0 \\ 1, & t > -t_0 \end{cases}$$



- The step function is used to represent an abrupt change in voltage or current in the circuit.

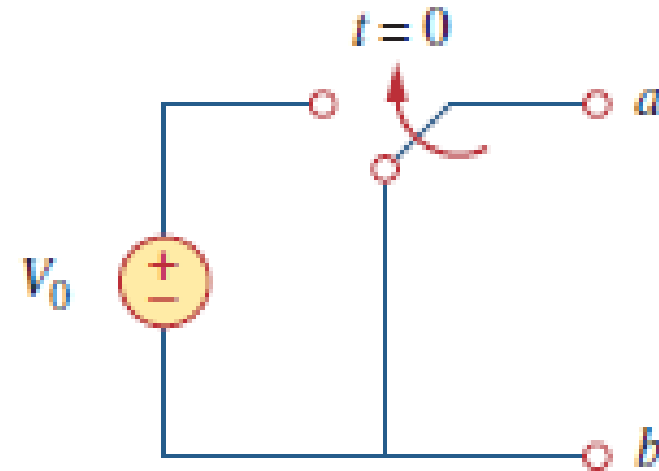
$$v(t) = \begin{cases} 0, & t < t_0 \\ V_0, & t > t_0 \end{cases}$$

$$v(t) = V_0 u(t - t_0)$$

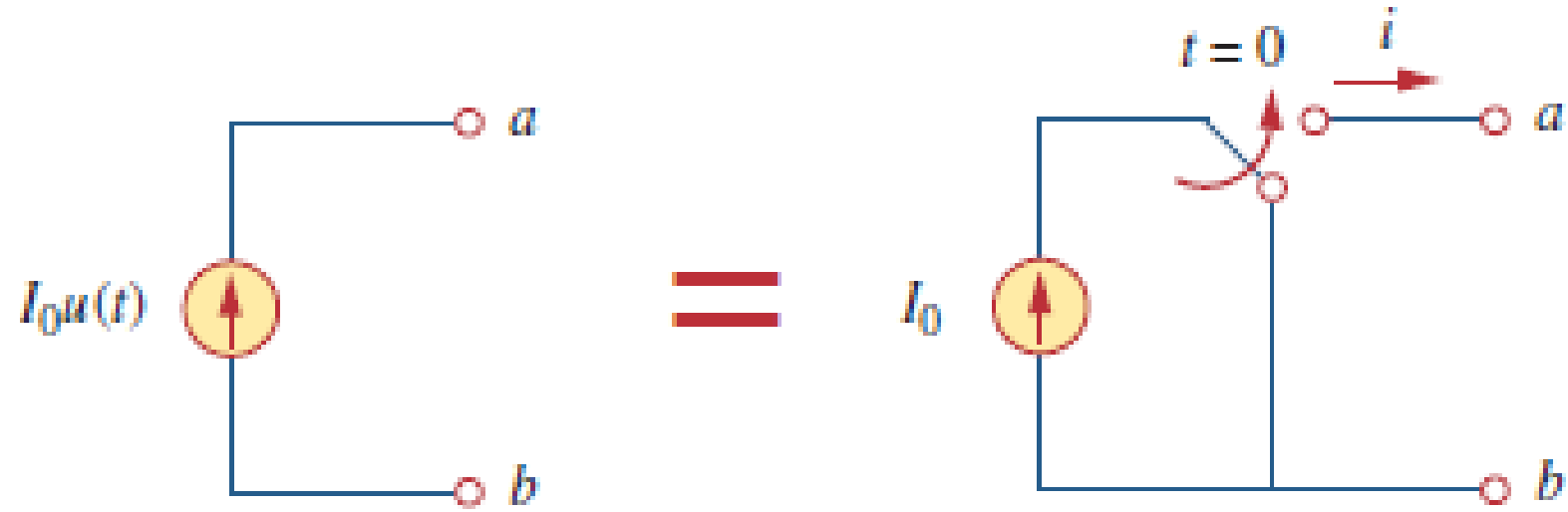


(a) Voltage source of $V_0 u(t)$.

=



(b) its equivalent circuit.



(a) Current source of $I_0 u(t)$.

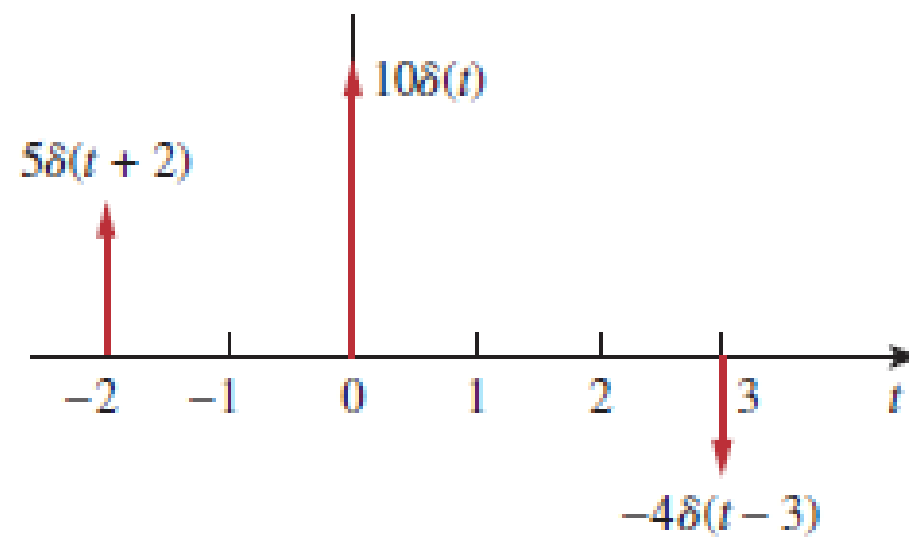
(b) its equivalent circuit.

The Unit Impulse function

- The derivative of the unit step function $u(t)$ is the unit impulse function.
- The unit impulse function is also known as delta function.

The unit impulse function $\delta(t)$ is zero everywhere except at $t = 0$, where it is undefined.

$$\delta(t) = \frac{d}{dt}u(t) = \begin{cases} 0, & t < 0 \\ \text{Undefined}, & t = 0 \\ 0, & t > 0 \end{cases}$$



$$\int_{0^-}^{0^+} \delta(t) dt = 1$$

$$\int_a^b f(t) \delta(t - t_0) dt = f(t_0)$$

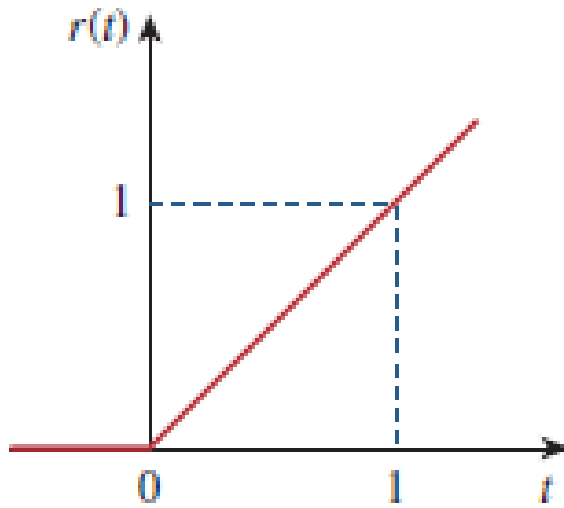
The Unit Ramp function

- Integrating the unit step function results in the *unit ramp function*

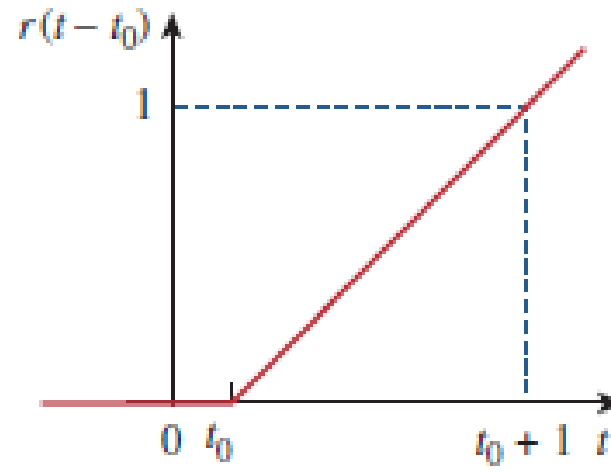
$$r(t) = \int_{-\infty}^t u(\lambda) d\lambda = tu(t)$$

$$r(t) = \begin{cases} 0, & t \leq 0 \\ t, & t \geq 0 \end{cases}$$

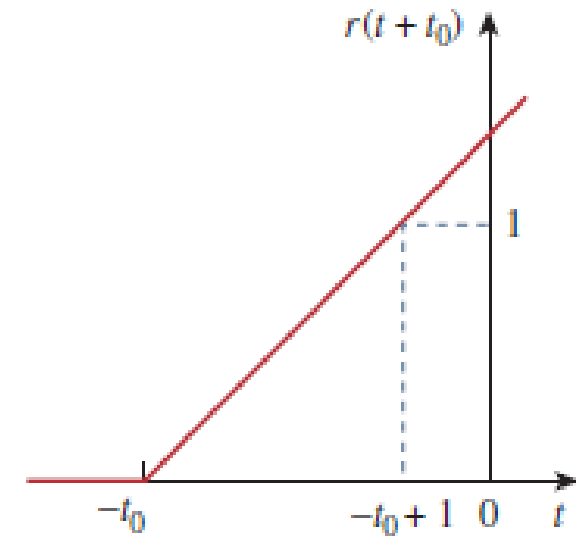
The unit ramp function is zero for negative values of t and has a unit slope for positive values of t .



$$r(t) = \begin{cases} 0, & t \leq 0 \\ t, & t \geq 0 \end{cases}$$



$$r(t - t_0) = \begin{cases} 0, & t \leq t_0 \\ t - t_0, & t \geq t_0 \end{cases}$$



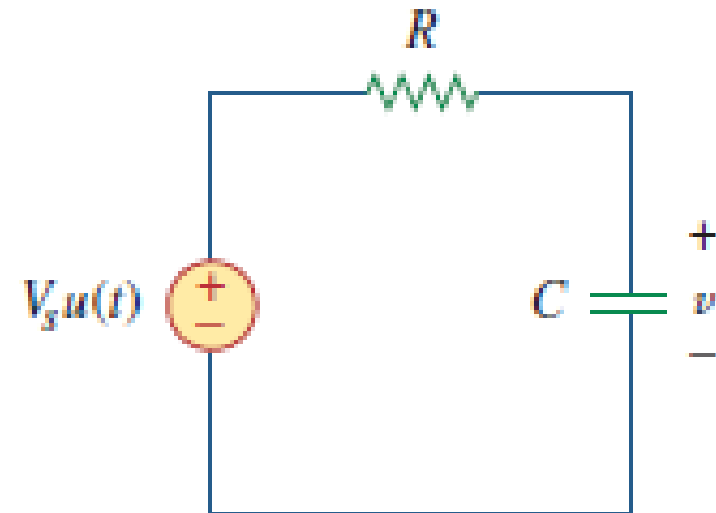
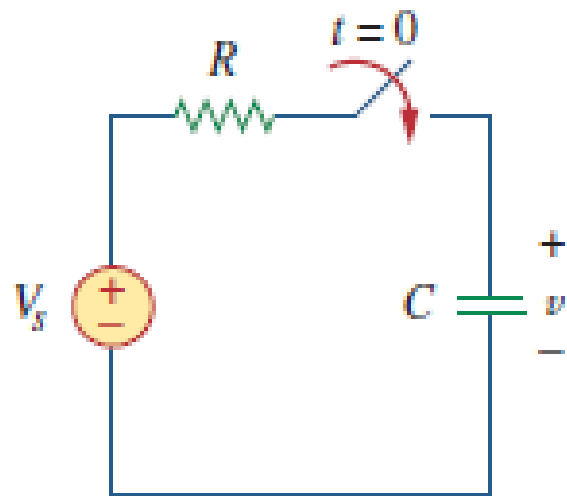
$$r(t + t_0) = \begin{cases} 0, & t \leq -t_0 \\ t + t_0, & t \geq -t_0 \end{cases}$$

Relationship between singularity functions:

$$\delta(t) = \frac{du(t)}{dt}, \quad u(t) = \frac{dr(t)}{dt} \qquad u(t) = \int_{-\infty}^t \delta(\lambda) d\lambda, \quad r(t) = \int_{-\infty}^t u(\lambda) d\lambda$$

Step response of an RC Circuit

The step response of a circuit is its behavior when the excitation is the step function, which may be a voltage or a current source.



$$v(0^-) = v(0^+) = V_0$$

$$C \frac{dv}{dt} + \frac{v - V_s u(t)}{R} = 0$$

$$\frac{dv}{dt} + \frac{v}{RC} = \frac{V_s}{RC} u(t)$$

$$\frac{dv}{dt} + \frac{v}{RC} = \frac{V_s}{RC}$$

$$\frac{dv}{dt} = -\frac{v - V_s}{RC}$$

$$\frac{dv}{v - V_s} = -\frac{dt}{RC}$$

Integrating both sides and introducing the initial conditions,

$$\ln(v - V_s) \Big|_{V_0}^{v(t)} = - \frac{t}{RC} \Big|_0^t$$

$$\ln(v(t) - V_s) - \ln(V_0 - V_s) = -\frac{t}{RC} + 0$$

Taking the exponential of both sides

$$\frac{v - V_s}{V_0 - V_s} = e^{-t/\tau}, \quad \tau = RC$$

$$v - V_s = (V_0 - V_s)e^{-t/\tau}$$

$$v(t) = V_s + (V_0 - V_s)e^{-t/\tau}, \quad t > 0$$

$$v(t) = \begin{cases} V_0, & t < 0 \\ V_s + (V_0 - V_s)e^{-t/\tau}, & t > 0 \end{cases}$$

Complete response of the circuit

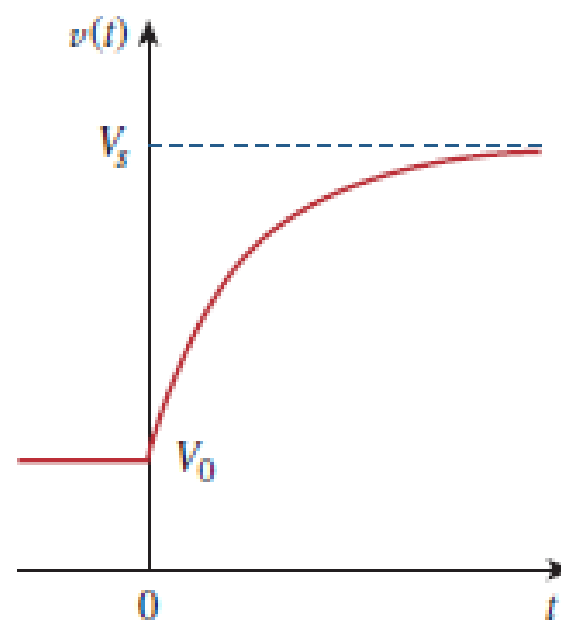
If we assume that the capacitor is uncharged initially,

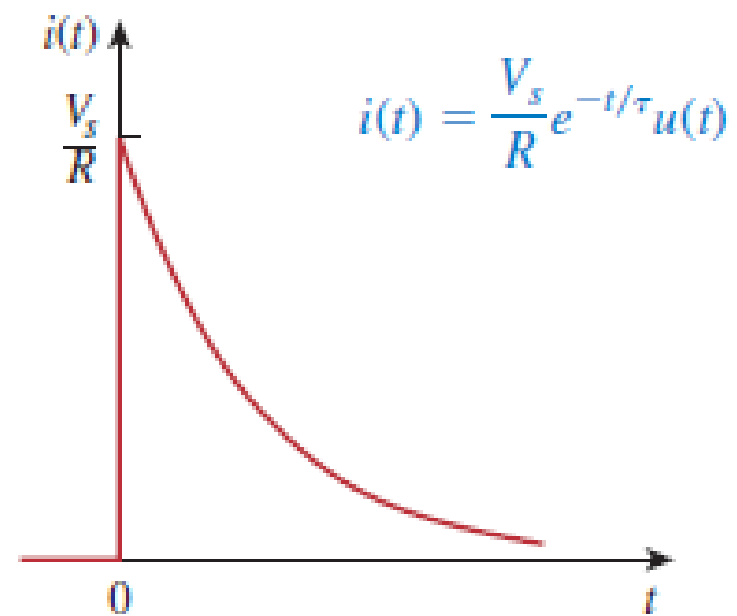
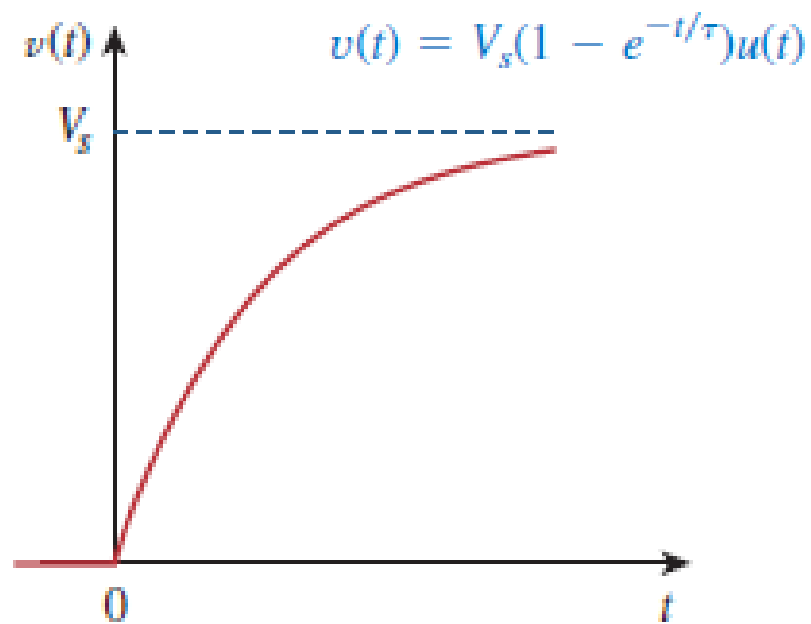
$$v(t) = \begin{cases} 0, & t < 0 \\ V_s(1 - e^{-t/\tau}), & t > 0 \end{cases}$$

$$v(t) = V_s(1 - e^{-t/\tau})u(t)$$

$$i(t) = C \frac{dv}{dt} = \frac{C}{\tau} V_s e^{-t/\tau}, \quad \tau = RC, \quad t > 0$$

$$i(t) = \frac{V_s}{R} e^{-t/\tau} u(t)$$





Step response of an RC circuit with initially uncharged capacitor: (a) voltage response, (b) current response.

Complete response = natural response + forced response
stored energy independent source

$$v = v_n + v_f$$

$$v_n = V_o e^{-t/\tau}$$

$$v_f = V_s(1 - e^{-t/\tau})$$

Complete response = transient response + steady-state response
temporary part permanent part

$$v = v_t + v_{ss}$$

$$v_t = (V_o - V_s)e^{-t/\tau}$$

$$v_{ss} = V_s$$

The **transient response** is the circuit's temporary response that will die out with time.

The **steady-state response** is the behavior of the circuit a long time after an external excitation is applied.

- Steps:

1. Find the initial capacitor voltage $v(0)$
2. Find the final capacitor voltage $v(\infty)$
3. Find the time constant

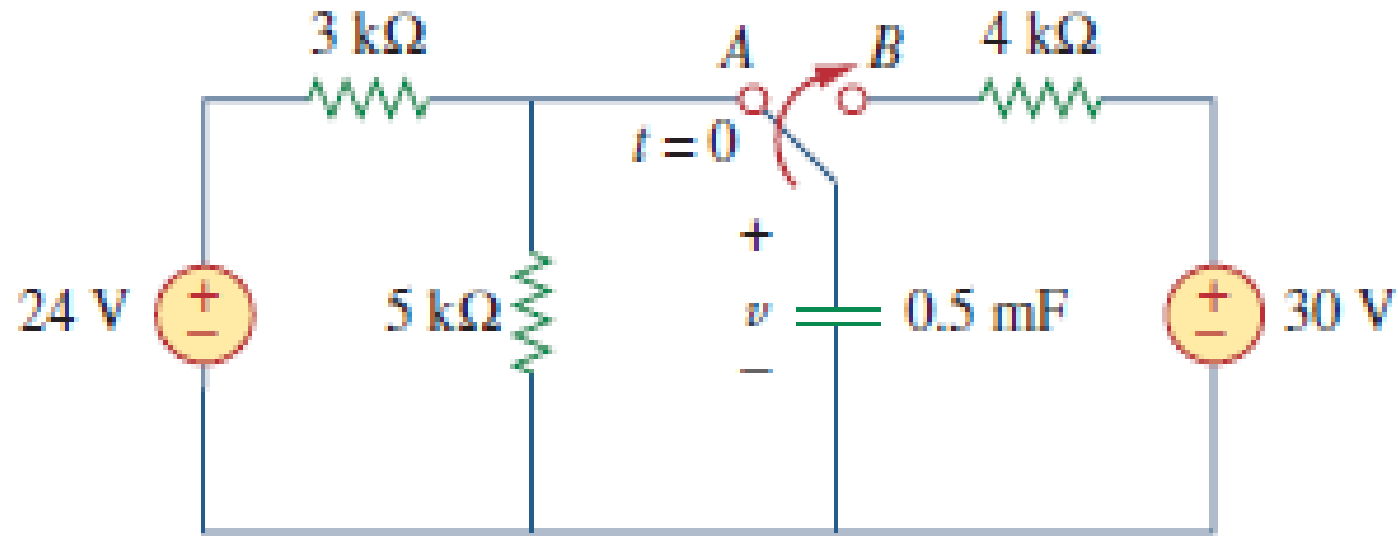
$$v(t) = v(\infty) + [v(0) - v(\infty)]e^{-t/\tau}$$

At time $t = t_0$

$$v(t) = v(\infty) + [v(t_0) - v(\infty)]e^{-(t-t_0)/\tau}$$

Numerical - 1

The switch has been in position A for a long time. At $t = 0$, the switch moves to B. Determine $v(t)$ for $t > 0$ and calculate its value at $t = 1$ s and 4 s.



Solution:

For $t < 0$, the switch is at position A . The capacitor acts like an open circuit to dc, but v is the same as the voltage across the $5\text{-k}\Omega$ resistor. Hence, the voltage across the capacitor just before $t = 0$ is obtained by voltage division as

$$v(0^-) = \frac{5}{5 + 3}(24) = 15 \text{ V}$$

Using the fact that the capacitor voltage cannot change instantaneously,

$$v(0) = v(0^-) = v(0^+) = 15 \text{ V}$$

For $t > 0$, the switch is in position B . The Thevenin resistance connected to the capacitor is $R_{\text{Th}} = 4 \text{ k}\Omega$, and the time constant is

$$\tau = R_{\text{Th}}C = 4 \times 10^3 \times 0.5 \times 10^{-3} = 2 \text{ s}$$

Since the capacitor acts like an open circuit to dc at steady state, $v(\infty) = 30$ V. Thus,

$$\begin{aligned} v(t) &= v(\infty) + [v(0) - v(\infty)]e^{-t/\tau} \\ &= 30 + (15 - 30)e^{-t/2} = (30 - 15e^{-0.5t}) \text{ V} \end{aligned}$$

At $t = 1$,

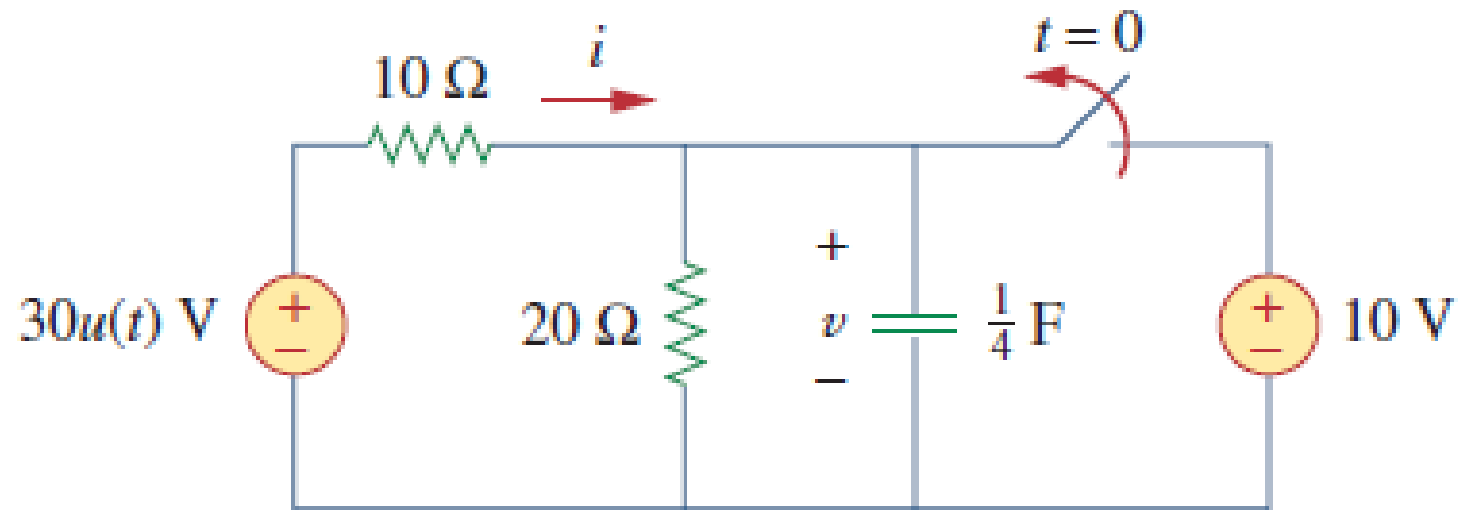
$$v(1) = 30 - 15e^{-0.5} = 20.9 \text{ V}$$

At $t = 4$,

$$v(4) = 30 - 15e^{-2} = 27.97 \text{ V}$$

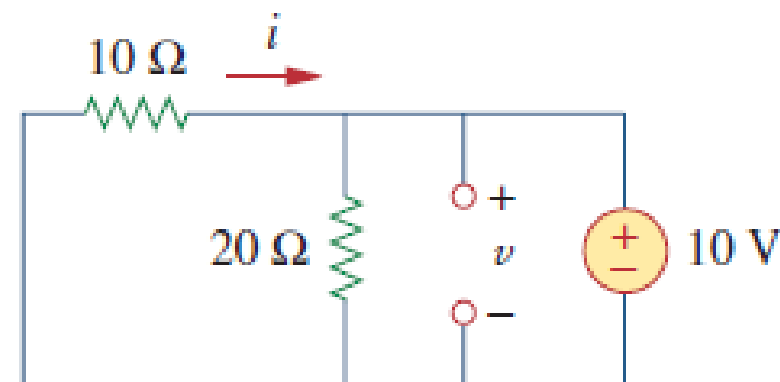
Numerical - 2

The switch has been closed for a long time and is opened at $t = 0$.
Find i and v for all time.



Solution:

$$30u(t) = \begin{cases} 0, & t < 0 \\ 30, & t > 0 \end{cases}$$



For $t < 0$, the switch is closed and $30u(t) = 0$,

$$v = 10 \text{ V}, \quad i = -\frac{v}{10} = -1 \text{ A}$$

Since the capacitor voltage cannot change instantaneously,

$$v(0) = v(0^-) = 10 \text{ V}$$

For $t > 0$, the switch is opened and the 10-V voltage source is disconnected from the circuit.

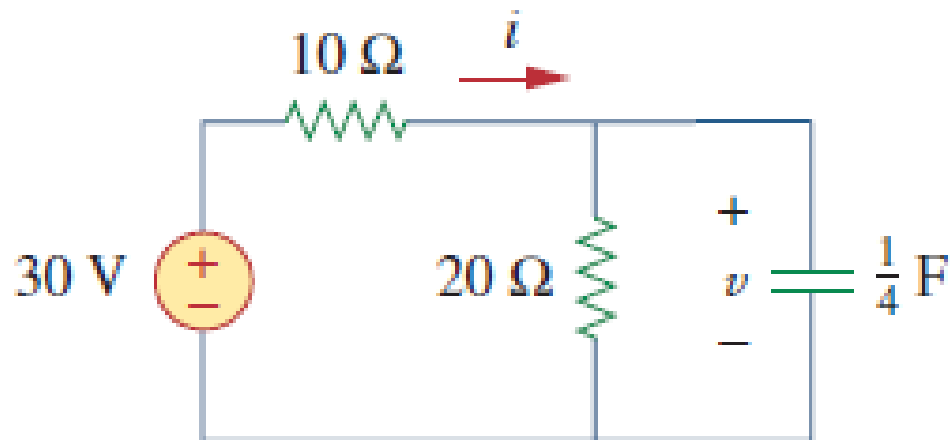
$$v(\infty) = \frac{20}{20 + 10}(30) = 20 \text{ V}$$

The Thevenin resistance at the capacitor terminals is

$$R_{\text{Th}} = 10 \parallel 20 = \frac{10 \times 20}{30} = \frac{20}{3} \Omega$$

$$\tau = R_{\text{Th}} C = \frac{20}{3} \cdot \frac{1}{4} = \frac{5}{3} \text{ s}$$

$$\begin{aligned} v(t) &= v(\infty) + [v(0) - v(\infty)]e^{-t/\tau} \\ &= 20 + (10 - 20)e^{-(3/5)t} = (20 - 10e^{-0.6t}) \text{ V} \end{aligned}$$



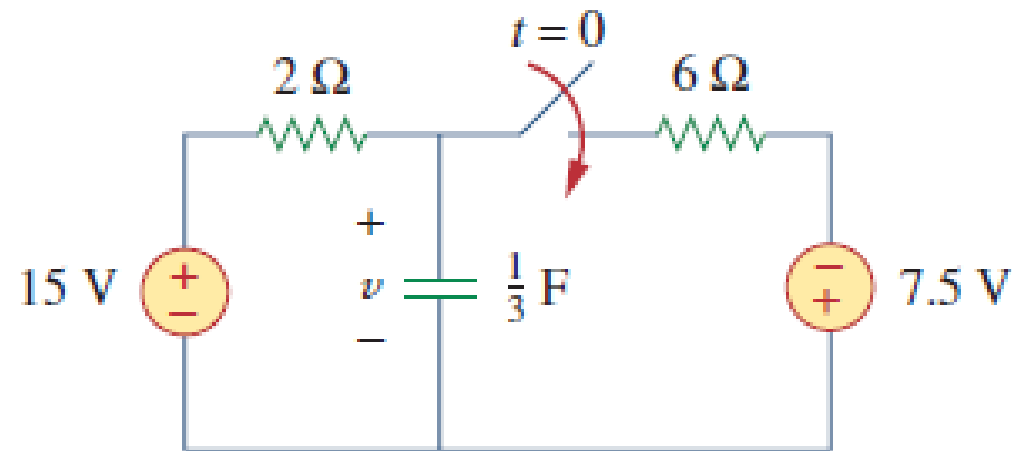
$$\begin{aligned}
 i &= \frac{v}{20} + C \frac{dv}{dt} \\
 &= 1 - 0.5e^{-0.6t} + 0.25(-0.6)(-10)e^{-0.6t} = (1 + e^{-0.6t}) \text{ A}
 \end{aligned}$$

$$v = \begin{cases} 10 \text{ V}, & t < 0 \\ (20 - 10e^{-0.6t}) \text{ V}, & t \geq 0 \end{cases}$$

$$i = \begin{cases} -1 \text{ A}, & t < 0 \\ (1 + e^{-0.6t}) \text{ A}, & t > 0 \end{cases}$$

Practice problem - 1

Find $v(t)$ for $t > 0$ in the circuit of Fig. Assume the switch has been open for a long time and is closed at $t = 0$. Calculate $v(t)$ at $t = 0.5$.



Answer: $(9.375 + 5.625e^{-2t})$ V for all $t > 0$, 7.63 V.